Unit 6 : Basic Network Management

6.1 IP address configuration: settings in a network. TCP/IP Network address, TCP/IP Configuration files, Network Interfaces and Routes: ifconfig, route, ping, netstat, tcpdump commands.
6.2 DHCP Configuration: Configuring DHCP Client and Server, Dynamic Address, Fixed Addresses.
6.3 NIS, NFS, SAMBA introduction.
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Network Intrusion Detection:

A network intrusion refers to any unauthorized activity on a digital network. Network intrusions often involve stealing valuable network resources and almost always jeopardize the security of networks and/or their data. In order to proactively detect and respond to network intrusions, organizations and their Cybersecurity teams need to have a thorough understanding of how network intrusions work and implement network intrusion, detection, and response systems that are designed with attack techniques and cover-up methods in mind.

Host Based Intrusion Detection:

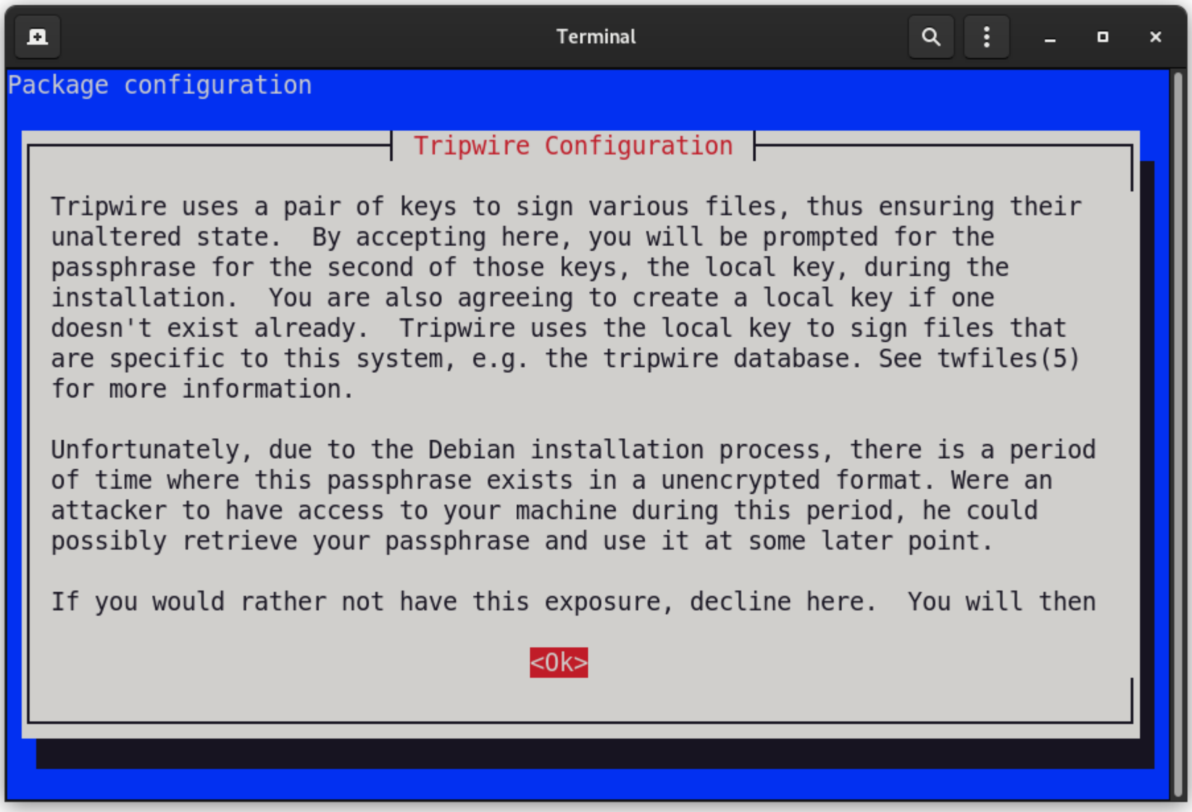
A host-based intrusion detection system is an intrusion detection system that is capable of monitoring and analyzing the internals of a computing system as well as the network packets on its network interfaces, similar to the way a network-based intrusion detection system operates.

Installing tripwire

Before we start, a very important piece of advice : try to install tripwire right after you install the system, because this way there are better chances it'll be clean, unaltered by malicious individuals. Tripwire creates a database of information related to your system, then compares that to what it finds when ran regularly, which it should, in order to get some real use out of it.

```
sudo apt-get install tripwire && tripwire --init
```

We say easy because the configure script asks you some basic configuration questions like system-wide passwords, so you get an easier start. dpkg- reconfigure will help you if something goes wrong and you want to reset. As you will see below, you will have to initialize tripwire's database, and this is applicable to every system tripwire is able to compile on.



Using Tripwire

```
$ tripwire -m i
```

Tripwire works by using modes. A mode is a function tripwire can execute, basically speaking. We already spoke of the first mode to use, the init mode. All tripwire modes can also be seen as actions, and every action-related flag (like --init) has a short equivalent, prefixed with -m. So, to initialize the database we could have written

One will obviously want to use tripwire after all this talking, so that can be done by using the check mode:

```
$ tripwire -m c
```

One flag you can use often in check mode is -I, which stands for interactive. You will find a huge number of problems found by tripwire when scanning, but don't panic. And of course, don't rely only on HIDS to check your system's integrity.

IDS software in general are known to generate false negatives/positives, hence the reports from such systems must be taken with a grain of salt. So, our check mode command becomes

```
$ tripwire -m c -I
```

Before we go on to database update mode, we must remind you to check the manual. Each mode has its specific options which you most likely will find useful, plus other options common to all or some of the modes, like -v, -c or -f (we invite you to find out what they do).

Since you will have to use these commands frequently, you should use cron or whatever tool you use for scheduling. For example, this line in root's crontab will do the trick:

```
45 04 * * * /usr/sbin/tripwire -m c
```

which will run the command daily at 04:45 AM.

In time, files on a system are changing. System updates, new installs, all these increase the discrepancies between the real thing and what tripwire knows about your system (the database).

Hence the database must be updated regularly in order to get reports as accurate as possible. We can easily accomplish this by typing

```
$ tripwire -m u
```

If you want to see the database in its' current form, twprint comes to the rescue:

```
$ twprint -m d
```

If you are even remotely involved in system security you will know what the term policy means. In tripwire terms, you define the policy in a file which will contain rules about which system object will be monitored, and how, to put it basically. '#' starts a comment, and the general rule for a line in the policy file is

#This is a comment and an example #

object -> property

/sbin -> $(ReadOnly)

! /data1

So, an object is basically a folder in your system, and here the second line shows how you should tell tripwire to leave the /data1 directory alone by using the '!' operator (C, anyone?). Regarding objects, take note that names like $HOME or ~ are never valid object identifiers and you will likely get an error message.

There are many things one should be aware of when writing or updating a policy file (rule attributes, variables, and so on), and tripwire looks promising and versatile in this respect.

You will find everything you can do with **tripwire's** policy file options in the manual page and some fine examples in /etc/tripwire/twpol.txt (at least on Debian systems).

twadmin will also be helpful when creating or checking configuration files or keys. For example, this command will print the policy file in its' current state:

```
$ twprint -m d
```

Finally, the test mode. What good is a monitoring tool if it can't report properly to you? This is what the test mode does. It e-mails the administrator, based on the settings found in the configuration file (first example) or as an command line option (second example) and if the mail is received correctly, life is good. This of course assumes your mail system is properly set up. Let's see:

```
$ tripwire -m t
$ tripwire -m t -e $user@$domain
```